

Amatya Sharma

Ph.D. Candidate in Computer Science and Engineering
University of Michigan, Ann Arbor

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EDUCATION

University of Michigan, Ann Arbor

Ph.D. Candidate in Computer Science and Engineering
Advisor: Prof. Nikhil Bansal and Euiwoong Lee.

Expected May 2027
GPA: 3.9/4.0

Indian Institute of Technology Kharagpur, India

Dual Degree: B.Tech. and M.Tech. in Computer Science and Engineering
Department Rank: 5; University Rank: 8.

May 2022
GPA: 9.69/10

RESEARCH INTERESTS

Approximation algorithms, streaming algorithms, parameterized algorithms, constraint satisfaction problems, clustering, computational geometry, online algorithms, and related problems in theoretical computer science.

PUBLICATIONS AND PREPRINTS

Authors are listed alphabetically by last name, as is standard in theoretical computer science.

- A. S. and Santhoshini Velusamy. **Non-Redundancy of Symmetric Boolean Predicates**. arXiv preprint 2026.
- A. S. and Santhoshini Velusamy. **Characterizing Streaming Decidability of CSPs via Non-Redundancy**. arXiv preprint 2026.
- Aditya Anand, Euiwoong Lee, Davide Mazzali, and A. S. **Min-CSPs on Complete Instances II: Polylogarithmic Approximation for Min-NAE-3-SAT**. APPROX 2025.
- Aditya Anand, Euiwoong Lee, and A. S. **Min-CSPs on Complete Instances**. SODA 2025.
- Nikhil Ayyadevara, Ashish Chiplunkar, and A. S. **A Decomposition Approach to the Weighted k -Server Problem**. FSTTCS 2024.
- Arindam Khan, Aditya Lonkar, Arnab Maiti, A. S., and Andreas Wiese. **Two-Dimensional Guillotine Strip Packing**. ICALP 2022.
- Arindam Khan, Arnab Maiti, A. S., and Andreas Wiese. **On Guillotine Separable Packings for the 2D Geometric Knapsack Problem**. SoCG 2021; HALG 2021.
- Palash Dey, Arnab Maiti, and A. S. **On Parameterized Complexity of Liquid Democracy**. CALDAM 2021.

RESEARCH EXPERIENCE

Research Visit, Toyota Technological Institute at Chicago

Host: Santhoshini Velusamy. Worked on streaming decidability and non-redundancy for constraint satisfaction problems.

Sep. 2025–Oct. 2025

Research Intern, Department of CSE, IIT Delhi

Mentor: Ashish Chiplunkar. Worked on online algorithms for the weighted k -server problem.

Apr. 2021–Oct. 2021

Research Intern, Department of CSA, IISc Bangalore

Mentor: Arindam Khan. Worked on approximation algorithms and computational geometry.

Dec. 2019

Research Intern, Department of CSA, IISc Bangalore

Mentor: Arindam Khan. Worked on approximation algorithms for geometric packing problems.

May 2019–Jul. 2019

TEACHING EXPERIENCE

Graduate Student Instructor, EECS 281: Data Structures & Algorithms, U of Michigan

Jan. 2026–Apr. 2026

Graduate Student Instructor, EECS 477: Intro to Algorithms, University of Michigan

Aug. 2024–Dec. 2024

Teaching Assistant, Algorithms-I, Department of CSE, IIT Kharagpur

Jan. 2022–Apr. 2022

Teaching Assistant, Algorithmic Game Theory, Department of CSE, IIT Kharagpur

Aug. 2021–Dec. 2021

MANUSCRIPTS AND ADDITIONAL RESEARCH PROJECTS

The Art Gallery Problem: A Survey
Dewang Modi, Adarsh Singh Parihar, A. S..

Manuscript

Surveyed algorithmic, geometric, and complexity-theoretic aspects of the Art Gallery Problem.

Parameterized Complexity of Nash Equilibrium of Networked Public Good Games

Manuscript

Palash Dey, A. S..

Studied parameterized complexity questions arising in networked public good games.

Parameterized Complexity of Margin of Victory

Manuscript

Palash Dey, A. S..

Studied parameterized complexity aspects of margin-of-victory problems in computational social choice.

Image Augmentation and Auxiliary Loss Duo

Manuscript

Faraaz Rehman Mallick, Dewang Modi, A. S..

Reproduced an AAAI 2021 paper on improving sample efficiency in model-free reinforcement learning.

Gaussian Process Kernels Survey

Manuscript

Prince Mahesh Bharadwaj, Adarsh Goyal, Nwe Ni Kyaw, A. S..

Surveyed local and global approximations and automated learning techniques for Gaussian process kernels.

INDUSTRY EXPERIENCE AND SOFTWARE PROJECTS

Intern, Oracle R&D, ExaCC Team

May 2021–Jul. 2021

Implemented a Java library for secure HTTP authentication using Java Cryptography Architecture.

Shoten

Jan. 2019–Jul. 2019

Built a web application using MySQL, JSP, HTML, and CSS to serve as an e-book store and print-request portal at institute level.

TinyC Compiler

Jul. 2019–Nov. 2019

Implemented parser and lexer components for a compiler for TinyC, a reduced subset of C, using Yacc, Bison, Flex, C, and C++.

Reduced Instruction Set Computer processor simulation (RISC) Processor

Jul. 2019–Nov. 2019

Designed a single-cycle executable processor for a subset of MIPS instructions and simulated it on FPGA boards using Verilog.

SKILLS AND GRADUATE COURSEWORK

Theoretical CS: Approximation and online algorithms; parameterized algorithms; randomized algorithms; algorithmic game theory; advanced graph theory; computational geometry; computational complexity; cryptography; selected topics in algorithms.

Learning Theory and AI: Reinforcement learning; deep learning; advanced machine learning; natural language processing; statistical foundations of AI and ML.

Software and Tools: Python; C/C++; Java; MySQL; Java Cryptography Architecture; MATLAB; HTML/CSS; JSP; Git.

Additional Coursework: Discrete mathematics; operating systems; networks; software engineering; linear algebra; probability and statistics.

ACADEMIC ACHIEVEMENTS, FELLOWSHIPS, AND GRANTS

Conference Grants and Travel Support

- APPROX 2025 Conference Grant, \$200 support to present at APPROX 2025.
- Rackham Conference Travel Funding, \$900 travel support to present at APPROX 2025.
- NSF Travel Support Grant, \$700 travel support to present at SODA 2025.
- Rackham Conference Travel Funding, \$1400 travel support to present at FSTTCS 2024.
- DIMACS Tutorial on Fine-Grained Complexity Travel Grant, travel support for the DIMACS tutorial.

Academic Achievements

- Department Rank 5 in the Department of Computer Science and Engineering, IIT Kharagpur.
- University Rank 8 at IIT Kharagpur.
- GATE Scholarship for teaching assistantship at the Department of Computer Science and Engineering, IIT Kharagpur.
- Department Change: moved to CS after ranking in the top 10/1300 students at IIT Kharagpur.
- JEE Advanced 2017: secured All India Rank 1464 among approximately 170,000 candidates.
- SJVN Merit Scholarship for performance in the CBSE class 12 examination.
- National Talent Search Examination: All-India top-1000/600,000; State Rank 1, Himachal Pradesh.
- Rashtriya Indian Military College Examination: secured a national rank in the top 60; State Rank 1.

SERVICE AND EXTRACURRICULAR ACTIVITIES

- **Conference Reviewing:** ICALP 2026; ITCS 2026; SODA 2026; APPROX 2024; ISAAC 2023; MFCS 2023.
- **Theory Talks and Lunches** (2023–2024): organized Theory CS group talks and lunches at the University of Michigan.
- **Annapurna** (2018–2019): participated in an initiative working against hunger and food wastage at IIT Kharagpur.